

AI4012 — Assignment #1 Report

Student: Haseeb Arshad (23i-2578) | **Task:** #81 Energy & Utilities / Grid — Outage status (GET /v1/energy/outages, {"region": "TX"})

Section 1 — System Architecture

The system answers one question for an AI agent: “Who is without power in Texas right now, where, how many, when is it coming back — and can I trust the number?” It is a Python / FastAPI / PostgreSQL pipeline with four moving parts.

1. Ingestion (the collector). A worker (app/ingestion.py) runs once or continuously (- - loop, every 5 min). Each pass asks **seven independent online sources**:

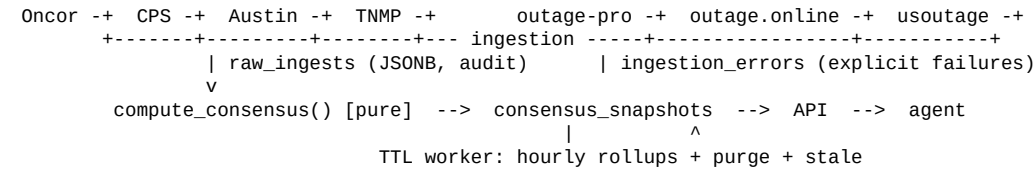
Source	Kind	What it gives
Oncor, CPS Energy, Austin Energy, TNMP (Kubra Storm Center JSON)	utility, first-hand	customers out by county, customers tracked, restoration ETA, own timestamp
outage-pro.com, outage.online, usoutage.com (HTML, scraped)	aggregator	customers out for all 254 counties and for 5–68 utilities

Every reply is saved to raw_ingests (JSONB) as an audit trail — JSON verbatim, HTML as the raw page (compressed) plus the extracted numbers. A timeout, block, server error or unrecognised page is written explicitly to ingestion_errors and named in the snapshot's warnings. Old data is never reused quietly and a failed fetch is never presented as a success.

2. Consensus (the judge). A pure function (app/consensus.py) turns the sources' numbers into one answer per county and per utility. Every reported value defines a candidate cluster (values within max(25, 20 %) of it); the cluster carrying the most reliability weight wins (utility feeds 0.55, aggregators 0.15), its members are averaged by weight and the rest are rejected as outliers. Confidence starts at 0.90–0.97 by in-cluster spread and falls in proportion to the weight rejected; **verified** needs ≥ 2 agreeing sources, confidence ≥ 0.85 and a weight majority. A lone source, or nobody agreeing, yields the most conservative (highest) claim at confidence 0.5, never verified. A utility's own county figure votes as one first-hand value, breaks 2-vs-2 splits, supplies the restoration **ETA**, and is a **floor**: we never serve fewer customers out than the utility itself admits. A **quality score** records how complete each source's record was.

3. Storage & lifecycle (the memory). PostgreSQL with indexed **JSONB**: raw_ingests (GIN on payload), consensus_snapshots (one row per pass; areas and utilities as JSONB with GIN indexes; provenance on the row so it survives purging), area_rollups, ingestion_errors, request_log. A **TTL worker** (app/ttl.py) summarises snapshots older than 24 h into hourly per-county rollups (samples, max, avg, min confidence) before deleting them, purges raw payloads older than 6 h, and flags snapshots older than the 15-minute freshness TTL as stale.

4. Serving (the counter). FastAPI (app/main.py) exposes GET /v1/energy/outages?region=TX (optional area, min_customers, limit). It reads **only the newest snapshot from Postgres** — never a live provider — so p95 latency stays far under 200 ms. It returns the standardised envelope: data (outages by county with customers affected, tracked, ETA, sources, confidence; utility-level figures; total; as_of) and meta (request id, product id, freshness with age/TTL/stale, provenance, trust, license, api, warnings). Unknown regions get a clean 404 error object; the advertised rate limit (100 / 60 s per client) is enforced with a clean 429. Every request is logged.



Section 2 — Use of AI (models and prompts)

Tool: Claude Code (Anthropic's agentic CLI), September 2026. **Model:** Claude Opus 5 (claude-opus-5).

How it was used. I used Claude Code as a pair-programming assistant against my written specification and the catalog row for task #81. It probed candidate online sources with me (PowerOutage.us blocks automated access and FEMA's EAGLE-I layer needs a token, so both were dropped; the four Kubra utility feeds and the three aggregator pages were confirmed live and keyless), then generated the module layout, the seven source adapters, the pure consensus function, the schema, the FastAPI layer, the TTL worker, the pytest suite and the GitHub Actions workflows. It ran everything in my environment — migration, live ingestion, the API, the TTL worker, the tests — and iterated on what the real data showed: the first consensus rule (median-based) collapsed on 2-vs-2 splits and had no tiebreaker when two aggregators disagreed, so it was replaced with weighted clustering plus the utility first-hand vote and floor; a third county-level aggregator was added after the first run showed only two; the outage-pro parser was fixed when rows with "<0.01 %" were found to be dropped. I directed the design decisions (sources, weights, tolerances, floor/ETA rules, TTLs, schema) and approved each step.

Representative prompts.

- "Above is the API I chose (#81, GET /v1/energy/outages, region TX) — redo everything in here please and make sure nothing is looked over in any manner at all."
- "Add more sources if you can please but using the exact API or task I want in here."
- "Make a GitHub workflow inside GitHub Actions for this repo if necessary since we are working with the API in a minute-wise manner."
- "Run the TTL worker once more and add the above results in a structured manner in res.md."

What I verified and changed myself.

- Confirmed all 40 tests pass and the served payload matches the required envelope exactly (data + meta: provenance, freshness, trust, license, api, warnings), including both 404 shapes.
- Verified live consensus on real data: Williamson County reported as 400 by Oncor's own map and outage-pro but 121 by two lagging aggregators — the first-hand vote resolved it to 400, verified, with the laggards listed as rejected; Harris County resolved by majority. Overall snapshot confidence 0.88, verified, quality 0.97.
- Verified the TTL worker rolls old snapshots into area_rollups (133 rows from one snapshot), purges raw rows and marks stale snapshots, with before/after row counts.
- Chose the consensus parameters (weights 0.55/0.15, tolerance max(25, 20 %), verified threshold 0.85, freshness TTL 15 min, raw TTL 6 h, snapshot TTL 24 h) and the seven-source lineup.

Limits of the AI's contribution. The AI did not choose the task, the trust model or the final source lineup; it had no access to the internet beyond the public pages the code fetches. All generated code was read and run by me before submission.

Appendix — Results of the final end-to-end run (15 Sep 2026)

Environment: Windows 10, Python 3.12, PostgreSQL 16.10 (native). Commands as in README.md, run in order. Full detail in docs/res.md.

A1. Live ingestion — all seven sources, ~15 s. Snapshot: total 11,539 customers affected, confidence 0.88, quality 0.97, verified. 0 source errors.

Source	Result	Customers out	Areas	Stamp (UTC)
ONCOR	ok	548	7 counties	07:10:00
CPS	ok	1	1	07:13:35
AUSTIN_ENERGY	ok	0	0 (no outages)	07:16:34
TNMP	ok	28	2	07:12:31
OUTAGE_PRO	ok	4,129	254	07:11:00
OUTAGE_ONLINE	ok	11,606	262	—
USOUTAGE	ok	12,478	261	06:27:05

A2. Consensus in action — real rows from this pass.

County	Reports (source → customers out)	Result	Conf.	Verified	Why
Harris	outage-pro 217 · outage.online 2,482 · usoutage 2,669	2,576	0.92	yes	majority cluster; outage-pro rejected
Williamson	Oncor 400 · outage-pro 400 · outage.online 121 · usoutage 121	400	0.93	yes	2-vs-2 split resolved by first-hand vote
Denton	Oncor 28 · usoutage 155 · outage.online 154	154	0.97	yes	lower first-hand = partial count, not outlier
Montgomery	three aggregators, none within tolerance	630	0.50	no	total disagreement → conservative claim
Oncor (utility)	Oncor 798 · outage-pro 798 · outage.online 1,593 · usoutage 2,017	798	0.93	yes	laggards rejected as outliers

A3. API. Envelope matched the Section 3 schema exactly.

Request	Status	Notes
GET /v1/energy/outages?region=TX&limit=8	200	total 11,539 · age 5 s · stale false · conf 0.88 · quality 0.97 · verified · latency 66 ms · 7 provenance entries
GET /v1/energy/outages?region=TX&area=Williamson	200	one row: 400 customers, 4 sources, conf 0.93
GET /v1/energy/outages?region=CA	404	{"error":{"code":"region_not_tracked",...},"meta":{"...}}

A4. TTL / lifecycle worker. The demonstration run summarised one snapshot into 133 hourly per-county rollup rows (276 rows over 143 counties in total) and deleted it; the normal run purged nothing because everything was younger than its TTL.

Run	Raw / snapshot TTL	Rollup rows	Snapshots purged	Raw purged	Marked stale
Normal (6 h / 24 h)	21600 s / 86400 s	0	0	0	0
Demonstration	60 s / 60 s	133	1	0	0

A5. Test suite — 41 passed, 0 failed, 0 skipped, 4.4 s.

File	Tests	Covers
tests/test_consensus.py	12	clustering, outlier rejection, first-hand tiebreak, floor, single source, total disagreement, failed-source warnings, quality
tests/test_sources.py	13	every parser on captured fixtures; home-county fallback; bad-schema errors; county-name unification
tests/test_api.py	9	exact envelope, filters, both 404 shapes, stale/fresh, unverified served honestly, rate limit 429
tests/test_sla.py	7	p95 < 200 ms (60 req), 200-request availability, failover ×3, TTL rollup+purge, stale marking

A6. Database state after the run.

Table	Contents after the final run
raw_ingests	7 rows (latest pass): utility JSON verbatim (1.6–58 kB); aggregator pages raw (gzip) + extracted data (~60 kB)
consensus_snapshots / area_rollups	1 fresh snapshot; 276 rollup rows over 143 counties (peak 2,576 in Harris)

Table	Contents after the final run
ingestion_errors / request_log	0 errors; 2,149 × HTTP 200 (avg 2 ms), 19 × HTTP 404 (avg 23 ms)